

Predicting Hungarian Definite vs. Indefinite Verb Conjugation from Sentence Context

Information Retrieval (5LN712), Uppsala University

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Links. GitHub: <https://github.com/elintove/hungarian-conjugation-classifier> Hugging Face dataset: <https://huggingface.co/datasets/elintove/hu-sentences-masked-verb> Demo: <https://huggingface.co/spaces/elintove/RagozoMeter>

1 Introduction

This project investigates whether sentence embeddings can support supervised classification for a Hungarian grammar-learning problem: deciding whether a masked transitive verb context requires definite or indefinite conjugation. The distinction is difficult for learners because it is not determined only by the presence of an object, but by the object’s definiteness and grammatical type. For example, a definite noun phrase such as *a könyvet* (Eng: *the book-ACC*) or a definite quantifier such as *mindkettőt* (Eng: *both-ACC*) requires definite conjugation, while universal pronouns such as *mindenkit* (Eng: *everyone-ACC*) and *mindent* (Eng: *everything-ACC*) trigger indefinite conjugation despite being semantically broad. The classification task is therefore domain-specific and linguistically meaningful: given a sentence containing [VERB], the model predicts *definite* or *indefinite*.

2 Dataset

A custom text classification dataset was generated with Python. The final version contains 1,200 masked Hungarian sentence contexts: 600 definite and 600 indefinite examples. The data is split into 960 training examples, 120 validation examples, and 120 test examples. Different sentence templates were used across train, validation, and test splits to reduce overfitting to one sentence structure. Because Hungarian is a pro-drop language, the subject distribution was controlled to imitate more natural input: 65% no explicit subject, 27.5% noun or proper-name subjects, and 7.5% pronoun subjects.

The dataset includes easy article-based cases, medium quantified noun phrases, and deliberately oversampled tricky categories such as definite quantifiers, demonstrative pronouns, third-person object pronouns, universal pronouns, and negative pronouns. This second 1,200-sentence version was created after the initial 800-sentence dataset showed weaker behavior on rare object types. This reflects a central data-design challenge: embeddings can capture useful semantic patterns, but the classifier still needs enough examples of the constructions that matter.

3 Method

Sentences were embedded with `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` [1, 4]. A multilingual sentence embedding model was selected because the input is Hungarian and the model directly produces sentence-level vectors, making it practical for a Hugging Face demo. Three classifiers were trained on the same embeddings: Logistic Regression, Linear SVM [2], and Random Forest [3]. This isolates the effect of the classifier while keeping the representation fixed. Models were evaluated with accuracy, macro F1, and per-class precision, recall, and F1. The best model was then used in a Gradio Space, where users enter a sentence with [VERB] and receive the predicted conjugation type with confidence scores. Additionally, a small object-type error analysis was carried out.

4 Results

Model	Class	800 sentences			1200 sentences		
		Prec	Rec	F1	Prec	Rec	F1
Logistic Regression	0	0.76	0.70	0.73	0.92	0.77	0.84
	1	0.72	0.76	0.75	0.80	0.93	0.86
Linear SVM	0	0.93	0.65	0.76	0.90	0.77	0.83
	1	0.73	0.95	0.83	0.80	0.92	0.85
Random Forest	0	0.82	0.80	0.81	0.98	0.70	0.82
	1	0.80	0.83	0.81	0.78	0.98	0.86

Table 1: Per-class performance at two dataset sizes. Class 0 is definite and class 1 is indefinite.

Model	800 sentences		1200 sentences	
	Accuracy	Macro F1	Accuracy	Macro F1
Logistic Regression	0.74	0.73	0.85	0.85
Linear SVM	0.80	0.80	0.84	0.84
Random Forest	0.81	0.81	0.84	0.84

Table 2: Overall model performance at two dataset sizes.

Increasing the dataset from 800 to 1,200 examples improved all models, especially Logistic Regression, which achieved the best final performance with 0.85 accuracy and 0.85 macro F1. Although Random Forest produced very high individual scores for class 0 precision and class 1 recall, it was less balanced overall, with lower recall for class 0. Logistic Regression was therefore selected for the final demo because it achieved the highest macro F1 and a more stable trade-off between the two classes.

5 Discussion and Reflection

The results suggest that multilingual sentence embeddings are useful for representing Hungarian sentence contexts in this controlled grammar task. However, error analysis showed that the model was less confident on unseen names and rare tricky constructions. This indicates that the dataset still contains some lexical memorization and should be expanded with more proper names, placeholder name examples, and more complex clauses. Future work could also compare the multilingual sentence-transformer with a Hungarian-specific transformer such as huBERT or PULI-BERT. Furthermore, the model was trained on single clause transitive sentences. It would be desirable to train a model on more complex sentences with multiple subclauses, to imitate more natural Hungarian.

The object-type analysis also explains why Logistic Regression was chosen despite Random Forest having some strong individual scores. Random Forest performed well on universal pronouns (0.90 accuracy) and third-person pronouns (0.86), but it performed poorly on definite articles (0.20) and definite quantifiers (0.50). Logistic Regression was more stable overall, with better performance on definite quantifiers (0.67) and a higher macro F1 score. This suggests that Random Forest overemphasized some object patterns, while Logistic Regression provided a better balance between the two main classes. Because some object-type groups are small, these values are used as qualitative error analysis rather than definitive category-level benchmarks.

AI tools were used for brainstorming ideas, generating reproducible dataset scripts, writing training code, understanding how to create a demo on HuggingFace and improving the report. The use of these tools were helpful and made the process easier, since they quickly helped solving bugs and problems that occurred along the way. It was inspiring to see how much could be done and modified from just a few prompts. At the same time, it did feel a bit fundamentally wrong, even though the instructions allowed it. It also made me doubt my own competence and knowledge, since much of what was done, I would not have been able to implement on my own without any use of AI tools. However, I will do my best to let it inspire me and learn from it instead. Furthermore, the process required manual checking of for example Hungarian grammar details, such as the behavior of *mindent* versus *mindet* and the natural omission of subjects had to be reviewed carefully. I used prompts to modify the dataset generation script, until I was happy with the grammar, subject/object distribution and naturalness of the sentence structures. I also asked a native Hungarian speaker to inspect some sentences I felt unsure about. Lastly, I would like to add that it was fun and useful to learn how to use GitHub and HuggingFace through this assignment.

References

- [1] Reimers, N., & Gurevych, I. (2019). Sentence-BERT: Sentence embeddings using Siamese BERT-networks. *EMNLP-IJCNLP*.
- [2] Cortes, C., & Vapnik, V. (1995). Support-vector networks. *Machine Learning*, 20, 273–297.
- [3] Breiman, L. (2001). Random forests. *Machine Learning*, 45, 5–32.
- [4] Hugging Face. (n.d.). *sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2*. <https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2>